

# Analysis 2: Lebesgue Integration and Fourier Series

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# 1 Mon January 12

Midterms 2/23 and 4/13, both in class.

## 1.1 Introduction

We use the **Lebesgue integral** for making sense of the integral when  $f$  is not a nice function. We use **Fourier series** for approximating periodic functions by the sum of sine waves. Historically, these two topics are actually intertwined!

Suppose we have a series of functions  $f_n : [a, b] \rightarrow \mathbb{R}$  all continuous and  $f_n \rightarrow f$  uniformly. Then we know  $f$  is continuous and the integrals converge.

Now suppose  $f_n$  Riemann integrable and  $0 \leq f_{n+1} \leq f_n$  and since bounded monotonic sequences converge, both

$$\lim_{n \rightarrow \infty} f_n(x)$$

and

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx$$

exist. But is it true that

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b \lim_{n \rightarrow \infty} f_n(x) dx?$$

For the Riemann Integral, it turns out the answer is no, because the limiting function might not be integrable.

### Example 1.1

Fix  $[a, b]$ . List all rationals in  $[a, b]$ :  $\{q_n\} = [a, b] \cap \mathbb{Q}$ . Then let

$$f_n(x) = \begin{cases} 0 & x \in \{q_1, \dots, q_n\} \\ 1 & \text{otherwise} \end{cases}$$

Here, each  $f_n$  is Riemann integrable because it only has finitely many discontinuities, but the limiting function  $f_n$  has infinitely many discontinuities, so it is not Riemann integrable. The Lebesgue integral fills in the hole by making the integral of the limit the limit of the integrals.

## 1.2 Riemann Integral

Riemann Integral was given by successive partitions and then lower sums and upper sums. Namely

$$U(P, f) = \sum_{k=1}^{n-1} (p_{k+1} - p_k) \sup(f)$$

where  $\sup(f)$  is on the partition. Lower sums increase and upper sums decrease upon refinements. When they are equal, we define the integral

$$\int = \sup_P L(P, f) = \inf_Q U(Q, f).$$

### Lemma 1.1

$f$  is Riemann integrable iff for every  $\varepsilon > 0$  there is a partition  $P$  of  $[a, b]$  such that  $U(P, f) - L(P, f) < \varepsilon$ .

*Proof.* ( $\implies$ ): Since  $\sup = \inf$ , for  $\varepsilon > 0$  we can find partitions  $Q$  and  $P$  such that  $U(Q, f) - L(P, f) < \varepsilon$ . Then let  $R = P \cup Q$  and

$$U(R, f) - L(R, f) \leq U(Q, f) - L(P, f) < \varepsilon.$$

( $\impliedby$ ): Given  $\varepsilon > 0$ , choose  $R$  such that  $U(R, f) - L(R, f) < \varepsilon$ . So  $\inf U(Q, f) - \sup L(P, f) < \varepsilon$ . Since  $\varepsilon$  is arbitrary,  $\sup = \inf$ .

This lets us prove

**Theorem 1.1**

If  $f$  is continuous, then  $f$  is Riemann integrable.

*Proof.* Check Real Analysis notes.

## 2 Wed Jan 14

Now to characterize integrability, we want to quantify discontinuity.

**Definition 2.1**

Given  $f : [a, b] \rightarrow \mathbb{R}$  and  $x \in [a, b]$ , let

$$\text{osc}(f, x) = \inf_{\delta > 0} \sup \{f(y) - f(z) \mid y, z \in [a, b] \text{ with } |y - x| < \delta \text{ and } |z - x| < \delta\}.$$

**Lemma 2.1**

$f$  is continuous iff  $\text{osc}(f, x) = 0$ .

*Proof.* ( $\implies$ ): For any  $\varepsilon > 0$ , use continuity to choose  $\delta > 0$  that implies  $|f(y) - f(x)| < \varepsilon$ . Then we can bound osc by  $2\varepsilon$ .

( $\impliedby$ ): Given  $\varepsilon > 0$ , use  $\text{osc} = 0$  to choose  $\delta > 0$  so  $\sup \leq \varepsilon$ , thus  $|f(y) - f(x)| \leq \varepsilon$  when  $|y - x| < \delta$ .

Some notation: a closed interval  $I$  has length  $|I|$  and interior  $\mathring{I} = (a, b)$ . Now we introduce our first notion of measure.

**Definition 2.2**

The **Jordan Content** of a bounded  $A \subseteq \mathbb{R}$  is

$$J(A) = \inf \{|I_1| + \cdots + |I_n| : I_1 \cup \cdots \cup I_n \supseteq A\}$$

- $J([a, b]) = b - a$
- For a finite set  $X$  of points,  $J(X) = 0$ , since we can make the  $I$ s as small as we want.

**Lemma 2.2**

If  $f : [a, b] \rightarrow \mathbb{R}$  is Riemann integrable, then  $f$  is bounded and for  $\varepsilon > 0$ ,  $J(\{x \mid \text{osc}(f, x) \geq \varepsilon\}) = 0$ . This is saying nearly the same thing as Riemann integrable functions have finitely many discontinuities.

*Proof.* Assume  $f$  is integrable, and fix  $\varepsilon > 0$ . For  $\delta > 0$  to be determined, choose a partition  $P$  so  $U(P, f) - L(P, f) < \delta$ . So we have finitely many intervals on which  $\sup - \inf$  of  $f$  is  $< \infty$ . Then write  $I_k = [p_k, p_{k+1}]$ . Call  $I_k$  “good” if

$$\sup_{I_k} f - \inf_{I_k} f < \varepsilon.$$

So if  $x \in \mathring{I}_k$  and  $I_k$  good,  $\text{osc}(f, x) < \varepsilon$ . Thus the bad intervals and the endpoints of all intervals cover  $\{x \mid \text{osc}(f, x) \geq \varepsilon\}$ . Observe

$$\begin{aligned} \delta &> U(P, f) - L(P, f) \\ &= \sum_k |I_k| \left( \sup_{I_k} f - \inf_{I_k} f \right) \\ &\geq \sum_{\text{bad}} \cdots \\ &\geq \sum_{\text{bad}} |I_k| \varepsilon \end{aligned}$$

so

$$\sum_{\text{bad}} |I_k| < \delta/\varepsilon.$$

Since we can choose  $\delta$  after  $\varepsilon$ , we can make this as small as we want, so

$$J(\{x \mid \text{osc}(f, x) \geq \varepsilon\}) = 0.$$

### Lemma 2.3

If  $f$  is bounded and for  $\varepsilon > 0$ ,  $J(\{x \mid \text{osc}(f, x) \geq \varepsilon\}) = 0$ , then  $f$  is Riemann integrable.

*Proof.*  $B = \sup_{[a,b]} |f| < \infty$ . Fix  $\varepsilon > 0$ . Choose  $I_1, \dots, I_n$  so  $\text{osc}(f, x) \geq \varepsilon \implies x \in I_1 \cup \dots \cup I_n$  and  $|I_1| + \dots + |I_n| < \varepsilon$ . We can fatten the intervals slightly so we get

$$\text{osc}(f, x) \geq \varepsilon \implies x \in \overset{\circ}{I}_1 \cup \dots \cup \overset{\circ}{I}_n.$$

We can also merge overlapping  $I$ s, and now assume all the  $I_k$  are disjoint.

Now if  $x \in [a, b] \setminus (\overset{\circ}{I}_1 \cup \dots \cup \overset{\circ}{I}_n)$ , then  $\text{osc}(f, x) < \varepsilon$ . Choose  $\delta_x$  such that  $\sup f - \inf f < \varepsilon$  on  $\tilde{I}_k = [a, b] \cap [x - \delta_x, x + \delta_x]$ . Since  $A = [a, b] \setminus (\overset{\circ}{I}_1 \cup \dots \cup \overset{\circ}{I}_n)$ , we can choose  $x_1, \dots, x_m$  so

$$A \subset (x_1 - \delta_{x_1}, x_1 + \delta_{x_1}) \cup \dots.$$

Now let  $P$  be the union of the endpoints of  $I_k$  and  $\tilde{I}_k$ . Each  $[p_k, p_{k+1}]$  is either contained in one of the  $I_j$  or  $\tilde{I}_j$ . Ok now we are introducing the normal intervals  $I'_k = [p_k, p_{k+1}]$ . Then

$$U(P, f) - L(P, f) = \sum_k |I'_k| (\sup_{I'_k} f - \inf_{I'_k} f).$$

Splitting this up into the bad and good intervals, the bad cases have are bounded by  $\varepsilon \cdot 2B$ , and the good ones are bounded by  $(b - a) \cdot \varepsilon$ . So both go to 0 as  $\varepsilon \rightarrow 0$ . Thus  $f$  is Riemann integrable.

### Theorem 2.1

$f : [a, b] \rightarrow \mathbb{R}$  is Riemann integrable iff  $f$  is bounded and  $\forall \varepsilon > 0$ ,  $J(\{x \mid \text{osc}(f, x) \geq \varepsilon\}) = 0$ .

But Jordan content is kind of difficult to work with. In particular, it doesn't play nice with infinite sums. So we introduce the Lebesgue measure.

### Definition 2.3

The **Lebesgue outer measure** of a set  $A \subset \mathbb{R}$  is

$$m^*(A) = \inf \left\{ \sum_{k=1}^{\infty} |I_k| : \bigcup_{k=1}^{\infty} I_k \supseteq A \right\}.$$

The Lebesgue measure of the rationals is 0, since if we let

$$I_k = [q_k - \varepsilon 2^{-k}, q_k + \varepsilon 2^{-k}]$$

the sum of all the lengths is  $2\varepsilon$ .

## 3 Wed Jan 21

### Lemma 3.1

$$J(A) \geq m^*(A).$$

*Proof.* We can convert a finite cover to an infinite cover by “padding”, adding intervals of length  $\varepsilon 2^{-k}$ , adding at most  $\varepsilon > 0$  total length.

**Lemma 3.2**

If  $K \subseteq \mathbb{R}$  is compact, then  $m^*(K) = J(K)$ .

*Proof.* We already know  $m^*(K) \leq J(K)$  from the previous lemma. Now fix  $\varepsilon > 0$ . Choose  $K \subseteq \bigcup_{k=1}^{\infty} I_k$  so  $\sum_{k=1}^{\infty} |I_k| \leq m^*(K) + \varepsilon$ . Let  $\hat{I}_k = (a_k - \varepsilon 2^{-k}, b_k + \varepsilon 2^{-k})$ . Since  $K$  is compact, we can create a finite subcover. We eventually get

$$\sum_{k=1}^N |\hat{I}_k| \leq 3\varepsilon + m^*(K).$$

**Lemma 3.3**

If  $A \subset B$ , then  $J(A) \subseteq J(B)$  and  $m^*(A) \leq m^*(B)$ .

*Proof.* Exercise. Should be simple? If  $I_k$  is an open cover of  $B$ , then it is an open cover of  $A$ .

**Lemma 3.4**

For  $A_k \subseteq \mathbb{R}$ ,

$$m^*\left(\bigcup_{k=1}^{\infty} A_k\right) \leq \sum_{k=1}^{\infty} m^*(A_k).$$

*Proof.* Dovetail it!

Fix  $\varepsilon > 0$ . For each  $k$ , choose cover for  $A_k$  that achieves  $m^*(A_k)$  with at most  $\varepsilon 2^{-k}$  error. Choose intervals  $I_{k,1}, \dots$  so

$$A_k \subseteq \bigcup_j I_{k,j}$$

and

$$\sum_{j=1}^{\infty} |I_{k,j}| \leq m^*(A_k) + \varepsilon 2^{-k}.$$

Let  $\{\hat{I}_l\} = \{I_{k,j}\}$ . Observe:

$$\bigcup_{k=1}^{\infty} A_k \subseteq \bigcup_{k=1}^{\infty} \bigcup_{j=1}^{\infty} I_{k,j} = \bigcup_{l=1}^{\infty} \hat{I}_l.$$

so

$$m^*\left(\bigcup_{k=1}^{\infty} A_k\right) \leq \sum_{l=1}^{\infty} |\hat{I}_l| \leq \sum_{k=1}^{\infty} (m^*(A_k) + \varepsilon 2^{-k}).$$

**Lemma 3.5**

If  $f : [a, b] \rightarrow \mathbb{R}$  and  $\varepsilon > 0$ , the

$$Z_{\varepsilon} = \{x \in [a, b] \mid \text{osc}(f, x) \geq \varepsilon\}$$

is compact.

*Proof.*  $Z_{\varepsilon}$  is bounded. So we just need to show it is also closed by showing it contains all its limit points.

For each  $n \geq 1$ , choose  $y_n$  and  $z_n \in [a, b]$  so  $|y_n - x_n| < 2^{-n}$  and same for  $z_n$ . Then  $f(y_n) - f(z_n) \geq \varepsilon - 2^{-n}$ . Using  $\text{osc}(f, x_n) \geq \varepsilon$ , since  $x_n \rightarrow x$ , we have  $y_n \rightarrow x$  and  $z_n \rightarrow x$ . Conclude  $\text{osc}(f, x) \geq \varepsilon$ .

**Theorem 3.1**

$f : [a, b] \rightarrow \mathbb{R}$  is Riemann integrable if and only if  $f$  is bounded and  $m^*(Z) = 0$  where  $Z = \{x \in [a, b] \mid f \text{ discontinuous at } x\}$ .

*Proof.* This follows from:

1.  $f : [a, b] \rightarrow \mathbb{R}$  is Riemann integrable iff  $f$  is bounded and, for  $n \geq 1$ ,  $J(Z_{1/n}) = 0$ .
2.  $Z = \bigcup_{n=1}^{\infty} Z_{1/n}$ .
3.  $m^*(Z) = 0$  iff for all  $n$   $J(Z_{1/n}) = 0$ .

## 4 Fri Jan 23

We continue the discussion of Cantor sets.

### Lemma 4.1

$C$  is nonempty and compact.

### Lemma 4.2

$$J(C_n) = \left(\frac{2}{3}\right)^n.$$

*Proof.* By induction, we have  $2^n$  disjoint intervals of length  $3^{-n}$ .

Let  $2^{\mathbb{N}}$  denote the set of sequences of 0 or 1. Given  $x \in 2^{\mathbb{N}}$ , define  $b(x) = \sum_{k=0}^{\infty} 2 \cdot 3^{-k-1} \cdot x_k$ .

### Lemma 4.3

$b : 2^{\mathbb{N}} \rightarrow C$  is a bijection.

*Proof.* The possible values of  $\sum_{k=0}^{n-1} 2 \cdot 3^{-k-1} \cdot x_k$  are the left endpoints of the intervals of  $C_n$ . Also  $0 \leq \sum_{k=0}^{\infty} 2 \cdot 3^{-k-1} \cdot x_k \leq 3^{-n}$ . Also  $b(2^{\mathbb{N}}) \subseteq C_n$ , and is dense in  $C$ . Also if  $x_n \neq \bar{x}_n$ , then  $|b(x) - b(\bar{x})| \geq 3^{-n}$ . This together somehow yields  $b$  is a bijection.

Let  $\mathbb{K}_C(x)$  be the indicator function of  $C$ . Then

### Lemma 4.4

$$\text{osc}(\mathbb{K}_C, x) = \mathbb{K}_C(x).$$

*Proof.* If  $x \notin C$ , then eis an open interval  $(a, b)$  with  $x \in (a, b)$  and  $(a, b) \cap C = \emptyset$ . So if  $x \notin C$ , then  $\text{osc}(\mathbb{K}_C, x) = 0$ . On the other hand, if  $x \in C$ , the complement of  $C$  is dense at  $x$ . Every neighborhood of  $x$  has a point  $y \notin C$ . So if  $x \in C$ , then  $\text{osc} = 1$ . Thus,  $\mathbb{K}_C$  is Riemann integrable.

## 4.1 Generalized Cantor Sets

Fix  $\tau : \mathbb{N} \rightarrow (0, 1)$ . Build Cantor set by removing middle  $\tau_n$ th fraction at step  $n$ .

### Lemma 4.5

$$J(C^\tau) = \prod_{n=1}^{\infty} (1 - \tau_n).$$

*Proof.* By induction, similar to the previous proof that  $J(C_n) = (2/3)^n$ .

## 4.2 Lebesgue Measure

We wish to replace the Riemann sum. Replace the lower sum with

$$\lim_{\delta \rightarrow 0} \sum_{k=1}^{\infty} \delta m^*({x \mid f(x) \geq k\delta}).$$

But we are missing countable additivity when  $A_k$  are disjoint, which is false in general.

## 5 Jan 28

Last time:

- $[a, b]$  measurable
- IF  $A$  and  $B$  are measurable, then so is  $A \cup B$ .
- If  $A$  measurable,  $\mathbb{R} \setminus A$  is measurable.
- If  $A_k$  measurable and disjoint, then  $\cup A_k$  is measurable and

$$m^*(\bigcup_k A_k) = \sum_k m^*(A_k).$$

Now onto today. We want to show

### Lemma 5.1

If an infinite set  $A_k$  are all measurable, then  $\bigcup_k A_k$  and  $\bigcap_k A_k$  are measurable.

*Proof.* Note we don't have  $A_k$  disjoint. Using 2 and 3, any finite combination of  $A_k$  is measurable. Basically we want to subtract all the previous  $A_1, \dots, A_{k-1}$  from  $A_k$  to create disjoint sets, which we know an infinite union of which is measurable.

### Lemma 5.2

If  $A_k$  measurable, then

$$m^*\left(\bigcup_{k=1}^{\infty} A_k\right) = \lim_{k \rightarrow \infty} m^*(A_1 \cup \dots \cup A_k).$$

And if we also have  $m^*(A_1) < \infty$ , then we can say the same for intersections.

### Lemma 5.3

Open and closed sets are measurable

*Proof.* Open sets are countable unions of intervals. And closed sets have open complements.

### Lemma 5.4

If  $A$  measurable and  $\varepsilon > 0$ , there are  $K \subseteq A \subseteq O$  with  $K$  closed,  $O$  open, and  $m^*(O \setminus K) < \varepsilon$ .

### Theorem 5.1

$A$  is measurable iff for every  $\varepsilon > 0$ , there are  $K \subseteq A \subseteq O$  with  $K$  closed,  $O$  open and  $m^*(O \setminus K) \leq \varepsilon$ .

## 6 Mon Feb 2

Vitali sets are not measurable, created with a construction using the axiom of choice. Define an equivalence relation  $\sim$  on  $\mathbb{R}$  such that  $a \sim b \iff a - b \in \mathbb{Q}$ . Using the axiom of choice, we choose one point from every equivalence class intersected with  $(1, -1)$ . This is a Vitali set  $E$ .

### Definition 6.1

If  $a \in B$  and  $B \subset \mathbb{R}$ , then

$$a + B = \{a + b \mid b \in B\}$$

is the translation of  $B$  by  $a$ .

Let  $q_k = \mathbb{Q} \cap (-2, 2)$ . We claim that  $q_k + E$  are disjoint, and

$$(-1, 1) \subseteq \bigcup_k (q_k + E) \subseteq (-3, 3).$$

*Proof.* (1): If there is an element in multiple  $q_k + E$ , then using the fact  $E$  only has one element from each equivalence class, we get  $q_K = q_j$ . (2):  $E \subseteq (-1, 1)$  and  $q_k \in (-2, 2)$ , so  $\bigcup_k (q_k + E) \subseteq (-3, 3)$ . Now supply  $a \in (-1, 1)$ . Since  $E$  contains an element from each equivalence class, there is an  $e \in E$  such that  $a - e \in \mathbb{Q} \cap (-2, 2)$ , there is  $k$  so  $a = q_k + e$ . Since  $a$  is arbitrary,  $(-1, 1) \subseteq \bigcup_k (q_k + E)$ .

### Lemma 6.1

Lebesgue outer measure is invariant under translation.

### Theorem 6.1

$A$  is not measurable if and only if there is a  $B \subseteq \mathbb{R}$  such that

$$m^*(B) < m^*(B \cap A) + m^*(B \cap A^c).$$

### Corollary 6.1

There is no function  $m : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$  such that

1.  $m([a, b]) = b - a$
2.  $A \subseteq B \implies m(A) \leq m(B)$
3.  $m(a + B) = m(B)$ .
4.  $A_k$  disjoint  $\implies m(\bigcup_k A_k) = \sum_k m(A_k)$ .

It turns out (3) is not necessary, and the above corollary is still true. We can use 1, 2, and 4 to prove regularity and then deduce (3).

We can just define the **Lebesgue Measure** as the restriction of the outer measure to only measurable sets.

### Proposition 6.1

$f : \mathbb{R} \rightarrow \mathbb{R}$  is measurable if and only if, for every open set  $O \subseteq \mathbb{R}$ ,  $f^{-1}(O) \subseteq \mathbb{R}$  is measurable.

## 7 Mon Feb 9

### Lemma 7.1

If  $f_n : \mathbb{R} \rightarrow \mathbb{R} \cup \{\pm\infty\}$  measurable, then  $\sup_n f_n$  and  $\inf_n f_n$  are measurable.

Now we can start with Lebesgue integration.

### Definition 7.1

An integrable simple function is a measurable function  $f : \mathbb{R} \rightarrow \mathbb{R}$  such that

- $f$  has finitely many values
- $m(\{x \mid f(x) \neq 0\}) < \infty$ .



Observe an integrable simple function can be uniquely written as

$$f = \sum_{k=1}^n a_k 1_{A_k}$$

with strictly increasing  $a_k$  and  $A_1, \dots, A_n \subseteq \mathbb{R}$  disjoint, measurable, and finite measure. Indeed, let  $\{a_k\} = f(\mathbb{R})$  and  $A_k = f^{-1}(\{a_k\})$ .

### Definition 7.2

The integral of an integrable simple function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is

$$\int f = \sum_{a \in f(\mathbb{R})} a m(f^{-1}(\{a\})).$$

For example, the indicator function on the Cantor set is an integrable simple function with integral  $m(C^T)$ .

### Lemma 7.2

If  $f, g$  are integrable simple functions and  $a \in \mathbb{R}$ , then

- $af$  is integrable and  $\int af = a \int f$
- $f + g$  is integrable and  $\int f + g = \int f + \int g$
- $f \leq g \implies \int f \leq \int g$ .

*Proof.* Exercise, but since the sums are finite it should behave as expected.

### Lemma 7.3

Suppose  $f : \mathbb{R} \rightarrow \mathbb{R}$  is measurable, and  $\sup_x |f(x)| < \infty$ , and has support with finite measure. For  $\varepsilon > 0$ , there are simple integrable  $g$  and  $h$  so  $g \leq f \leq h$  and  $h - g \leq \varepsilon$ .

*Proof.* Let  $g = \varepsilon \lfloor f/\varepsilon \rfloor$  and  $h = \varepsilon \lceil f/\varepsilon \rceil$ .

### Definition 7.3

If  $f : \mathbb{R} \rightarrow [0, +\infty]$  measurable, then

$$\int f = \sup\{ \int g : 0 \leq g \leq f, \quad g \text{ simple integrable} \}.$$

We want these definitions to agree for simple integrable functions! Indeed, they agree by Lemma 7.2. We also want to prove Lemma 7.2 for the integral on measurable functions.

### Definition 7.4

A measurable  $f : \mathbb{R} \rightarrow \mathbb{R} \cup \{\pm\infty\}$  is integrable if and only if  $\int |f| < \infty$ . When this happens, define  $\int f = \int f^+ - \int f^-$ .

## 8 Wed Feb 11

Last time: the Lebesgue integral.

Limit Theorems. We know if  $f_n$  measurable and  $f_n \rightarrow f$  pointwise, then  $f$  measurable. But this is false when we replace measurable by Riemann integrable.

In the monotone case, we have continuity of  $m$ :

$$A_n \subseteq A_{n+1} \implies m\left(\bigcup_n A_n\right) = \lim m(A_n)$$

and same for  $A_n \supseteq A_{n+1}$  if  $m(A_1) < \infty$ .

If  $f_n : \mathbb{R} \rightarrow [0, +\infty]$  measurable and  $f_n \leq f_{n+1}$ , then  $\int \lim_n f_n = \lim_n \int f_n$ . Here  $\lim f_n$  is pointwise limit which exists by monotonicity.

*Proof.* Use continuity of  $m$ .

### Proposition 8.1 Monotone Convergence II

If  $f_n : \mathbb{R} \rightarrow [0, +\infty]$  measurable,  $f_n \geq f_{n+1}$ , and  $\int f_1 < \infty$ , then  $\int \lim_n f_n = \lim_n \int f_n$ .

*Proof.*  $\int f_1 - \int \lim_n f_n$ .

### Lemma 8.1 Fatov's Lemma

If  $f_n : \mathbb{R} \rightarrow [0, +\infty]$  measurably then  $\int \liminf_n f_n \leq \liminf_n \int f_n$ .

*Proof.* Recall  $\liminf_n f_n(x) = \sup_n (\inf_{k \geq n} f_k(x))$ . Let  $g_n = \inf_{k \geq n} f_k$ . Observe  $g_n \leq g_{n+1} \rightarrow \liminf_n f_n$ . By monotone convergence,  $\lim_n \int g_n = \int \liminf_n f_n$ . Also,  $g_n \leq f_n \implies \int g_n \leq \int f_n$  and  $\liminf_n \int g_n \leq \liminf_n \int f_n$ .

**Dominated Convergence** If  $g, f_n : \mathbb{R} \rightarrow \mathbb{R} \cup \{\pm\infty\}$  measurable,  $|f_n| \leq |g|$ ,  $\int |g| < \infty$ , and  $f_n$  converges pointwise, then  $\lim_n \int f_n = \int \lim_n f_n$  and  $\lim_n \int |f_n - f| = 0$ .

*Proof.* We apply Fatov's lemma to  $2|g| - |f - f_n|$ .

## 9 Mon Feb 16

There are three types of convergence:

1.  $f_n \rightarrow f$  uniformly
2.  $f_n \rightarrow f$  pointwise
3.  $\int |f - f_n| \rightarrow 0$ .

We know  $1 \implies 2$  and  $1 \implies 3$ .

A **null set** is measurable  $A \subseteq \mathbb{R}$  with  $m(A) = 0$ . So any subset of a null set is a null set, and a countable union of null sets is null. We say  $f, g : \mathbb{R} \rightarrow \mathbb{R} \cup \{\pm\infty\}$  are equal almost everywhere if and only if  $\{f \neq g\}$  is null.

### Lemma 9.1

If  $f : \mathbb{R} \rightarrow [0, +\infty]$  measurable then  $\int f = 0$  iff  $f = 0$  almost everywhere.

If  $f$  continuous, then we must have  $f = 0$  everywhere.

### Lemma 9.2

If  $f, g : \mathbb{R} \rightarrow \mathbb{R} \cup \{\pm\infty\}$ , and they are equal almost everywhere, then  $f$  measurable implies  $g$  measurable.

### Lemma 9.3

If  $f : \mathbb{R} \rightarrow [0, +\infty]$  and  $\int f < \infty$ , then  $f$  is finite almost everywhere.

**Theorem 9.1**

If  $f : [a, b] \rightarrow \mathbb{R}$  is Riemann integrable, and extend  $f : \mathbb{R} \rightarrow \mathbb{R}$  by setting  $f(x) = 0$  for  $x \notin [a, b]$ , then  $f$  is Lebesgue integral and the integrals are equal.

**Lemma 9.4**

Suppose  $f_n, f : \mathbb{R} \rightarrow \mathbb{R} \cup \{\pm\infty\}$  is measurable and  $\lim_{n \rightarrow \infty} |f_n - f| = 0$ . There is a sequences  $n_k \rightarrow \infty$  and a null set  $A$  such that  $\lim_{k \rightarrow \infty} f_{n_k}(x) = f(x)$  for  $x \in A^c$ .

**9.1 Spaces of Functions**

Let  $\mathcal{L}$  denote the space of integrable functions  $f : \mathbb{R} \rightarrow \mathbb{R} \cup \{\pm\infty\}$ . Let  $\|f\| = \int |f| < \infty$ . We have the following norm axioms:

- $\|f\| \geq 0$
- $\|f\| = 0 \implies f = 0$
- $\|tf\| = t\|f\|$  for  $t > 0$
- $\|f + g\| \leq \|f\| + \|g\|$

but (2) fails for our norm. We will actually fix this by defining a new equivalence relation where  $f \sim g \iff f = g$  almost everywhere. But for now, we call our norm a seminorm. We have two main results

- $\mathcal{L}$  is complete: every Cauchy sequence has a limit
- Continuous and compactly supported functions are dense.

Thus the Lebesgue integral is the completion of the Riemann integral on the space of compactly supported continuous functions.

**10 Web Feb 18**

Didn't catch the first part of class.

Let  $C_C(\mathbb{R})$  be the space of continuous and compactly supported functions. That is,  $f \in C_C(\mathbb{R})$  if and only if  $f$  is a continuous map from  $\mathbb{R} \rightarrow \mathbb{R}$  and the closure of  $\{f \neq 0\}$  is compact.

If  $f \in C_C(\mathbb{R})$ , then both Riemann and Lebesgue integrals of  $f$  are defined and agree.

**Theorem 10.1**

$C_C(\mathbb{R})$  are dense in  $\mathcal{L}$ .

*Proof.* Suppose  $f = 1_A$  and  $m(A) < \infty$ . Then approximate  $A$  with  $K$  and  $O$ . Let  $r = \inf\{|x - y| \mid x \in K, y \in O\}$ . Exercise:  $r > 0$ . Then  $g = \max(0, 1 - \frac{1}{r} \min_{y \in K} |x - y|)$ . So  $1_K \leq g \leq 1_O$ . We compute

$$\begin{aligned} \int 1_g - 1_A &\leq \int 1_O - 1_K \\ &\leq m(O \setminus K) \leq \varepsilon. \end{aligned}$$

Now suppose  $f \geq 0$  and integrable. Choose integrable simple  $h$  such that  $0 \leq h \leq f$  and  $\int |f - h| \leq \varepsilon$ . Write  $h = \sum_{k=1}^n a_k 1_{A_k}$ . By what we proved above, choose  $g_k \in C_C(\mathbb{R})$  so  $|g_k - a_k 1_{A_k}| \leq \varepsilon 2^{-k}$ . Let  $g = \sum_{k=1}^n g_k$  and complete  $|g - h| \leq 2\varepsilon$ . Then by triangle inequality,  $|f - g| \leq 3\varepsilon$ .

In the most general case if  $f$  integrable, write  $f = f^+ - f^-$  and apply the previous case.

### Theorem 10.2

Suppose  $M$  is a metric space. There is a metric space  $\bar{M}$  and isometry  $i : M \rightarrow \bar{M}$  so that for any complete metric space  $\bar{N}$  and uniformly continuous  $f : M \rightarrow \bar{N}$  there is a unique uniformly continuous  $\bar{f} : \bar{M} \rightarrow \bar{N}$  so that  $\bar{f} \circ i = f$ .

For example, if  $f : \mathbb{Q} \rightarrow \mathbb{R}$  is uniformly continuous, then  $f$  has a unique extension  $\bar{f} : \mathbb{R} \rightarrow \mathbb{R}$ . Let  $L^1$  be the equivalence classes of  $\mathcal{L}^\infty$  with equivalence relation  $f \sim g$  iff  $f = g$  almost everywhere.

### Corollary 10.1

$L^1$  is the completion of  $C_C(\mathbb{R})$  equipped with  $\|f\| = \int |f|$ .

## 11 Web Feb 25

Today, stuff about completion, and  $L^1$  and  $L^2$ .

### Theorem 11.1

If  $M$  is a metric space, there is a complete metric space  $\bar{M}$  and an isometry  $i : M \rightarrow \bar{M}$  with the following “completeness” property: if  $\bar{N}$  is complete and  $f : M \rightarrow \bar{N}$  is uniformly continuous, then there is a unique uniformly continuous  $\bar{f} : \bar{M} \rightarrow \bar{N}$  so  $f = \bar{f} \circ i$ .

*Proof.* We know a metric is complete iff every Cauchy sequence converges. We say two Cauchy sequences  $\{a_n\}$  and  $\{b_n\}$  are equivalent if  $\lim_{n \rightarrow \infty} d(a_n, b_n) = 0$ . This is an equivalence relation by the triangle inequality. Let  $\bar{M}$  be the set of equivalence classes of Cauchy sequences. Check  $\bar{M}$  completes the metric space, then diagonalize. Let  $i(a) = [\{a\}]$  where  $\{a\}$  is a constant sequence. If  $f : M \rightarrow \bar{N}$  uniformly continuous and  $\{a_n\} \subseteq M$ , then  $\{f(a_n)\}$  Cauchy and set  $\bar{f}([\{a_n\}]) = \lim_n f(a_n)$ . Check  $\bar{f}$  unique.

Basically this is how we construct  $\mathbb{R}$ .

For  $f, g \in \mathcal{L}^1$ , write  $f \sim g$  iff  $\|f - g\| = 0$ . Let  $L^1$  be equivalence classes of  $\mathcal{L}^1$ . We can define  $i : C_C(\mathbb{R}) \rightarrow L^1$  by  $i(f) = [f]$ .

### Theorem 11.2

This  $i$  defined above is a completion of  $C_C(\mathbb{R})$ .

*Proof.* Since  $\|[f]\| = 0 \implies f \sim 0$ , we obtain  $L^1$  is a normed space.  $L^1$  is complete, since  $\{[f_n]\}$  Cauchy in  $L^1$  implies  $\{f_n\}$  is Cauchy in  $\mathcal{L}^1$  so  $\{f_n\} \rightarrow f$  in  $\mathcal{L}^1$  so  $[f_n] \rightarrow [f]$  in  $L^1$ .

We WTS that if  $F : C_C(\mathbb{R}) \rightarrow \bar{N}$  uniformly continuously, then there is a unique uniformly continuous  $\bar{F} : L^1 \rightarrow \bar{N}$  so  $F = \bar{F} \circ i$ . Suppose  $[f] \in L^1$ . By density, choose  $g_n \in C_C(\mathbb{R})$  so

$$\|g_n - f\| \leq 2^{-n}.$$

So  $g_n$  Cauchy. Unfortunately I can't read the rest so assume we proved this.

From now on, instead of writing  $[f] \in L^1$ , we write  $f \in L^1$ . When we say  $f = g$ , we mean  $f = g$  almost everywhere.

### 11.1 Square Integrable Functions

#### Definition 11.1

Let  $L^2$  be the space of (equivalence classes of) measurable functions from  $\mathbb{R} \rightarrow \mathbb{R} \cup \{\pm\infty\}$  such that  $\int f^2 < \infty$ .

If  $f, g \in L^2$ , then we compute by AM-GM.

$$\left| \int fg \right| \leq \int |fg| \leq \int f^2/2 + \int g^2/2 < \infty.$$

Technically we have to split the integrals into  $f_+$  and  $f_-$  and  $g_+$  and  $g_-$ , but I don't want to write that out.

Define

$$\langle f, g \rangle_2 = \int fg.$$

Observe

- $0 \leq \langle f, f \rangle < \infty$
- $\langle f, f \rangle = 0 \implies f = 0$
- $\langle f, g \rangle = \langle g, f \rangle$
- $\langle af + bg, h \rangle = a \langle f, h \rangle + b \langle g, h \rangle$

We can define the norm

$$\|f\|_2 = \sqrt{\langle f, f \rangle}.$$

Cauchy-Schwartz says

$$\langle f, g \rangle \leq \|f\|_2 \|g\|_2$$

with proof by AM-GM and writing  $ab = \sqrt{t} \cdot \frac{b}{\sqrt{t}}$  for  $t > 0$ .

### Theorem 11.3

$L^2$  is complete

*Proof.* Suppose  $f_n \in L^2$  Cauchy. Pass to subsequence so that

$$\|f_{n+1} - f_n\| \leq 2^{-n}.$$

For  $a < b$ , estimate

$$\int 1_{[a,b]} |f_{n+1} - f_n| \leq \|1_{[a,b]}\| \|f_{n+1} - f_n\| \leq \sqrt{b-a} \cdot 2^{-n}$$

by Cauchy-Schwartz. So  $1_{[a,b]} f_n$  is Cauchy in  $L^1$ . So  $\lim_n (1_{[a,b]} f_n)(x)$  exists for almost every  $x$  by proof of  $L^1$  completeness. Since  $[a, b]$  arbitrary, we have  $f(x) = \lim_n f_n(x)$  exists for almost every  $x$ . Compute

$$\|f - f_n\|_2^2 = \int (f - f_n)^2$$

and keep bounding it with clever sum tricks and we get a limit